

Equivalence Relations

Based on Ghorpade & Limaye

What is a Relation?

A **relation** R on a set S is a subset of $S \times S$.
We write $a \sim b$ if $(a, b) \in R$.

What is a Relation?

A **relation** R on a set S is a subset of $S \times S$.

We write $a \sim b$ if $(a, b) \in R$.

A relation $\sim$ is called an **equivalence relation** if it satisfies:

- 1 Reflexivity: $a \sim a$
- 2 Symmetry: $a \sim b \Rightarrow b \sim a$
- 3 Transitivity: $a \sim b$ and $b \sim c \Rightarrow a \sim c$

Examples of Equivalence Relations

- Equality on any set
- Congruence modulo n on $\mathbb{Z}$:

$$a \equiv b \pmod{n} \iff n \mid (a - b)$$

- “Has the same birthday as” on the set of people

Equivalence Class

Given an equivalence relation $\sim$ on S , the equivalence class of $a \in S$ is:

$$[a] = \{x \in S : x \sim a\}$$

Equivalence Class

Given an equivalence relation $\sim$ on S , the equivalence class of $a \in S$ is:

$$[a] = \{x \in S : x \sim a\}$$

Two elements $a, b \in S$ are equivalent if and only if $[a] = [b]$.

Equivalence Class

Given an equivalence relation $\sim$ on S , the equivalence class of $a \in S$ is:

$$[a] = \{x \in S : x \sim a\}$$

Two elements $a, b \in S$ are equivalent if and only if $[a] = [b]$.

If $a \not\sim b$, then $[a] \cap [b] = \emptyset$.

Partition of a Set

- A collection $\{A_i\}$ of subsets of S is a **partition** if:
 - Each A_i is nonempty,
 - $A_i \cap A_j = \emptyset$ for $i \neq j$,
 - $\bigcup A_i = S$

Partition of a Set

- A collection $\{A_i\}$ of subsets of S is a **partition** if:
 - Each A_i is nonempty,
 - $A_i \cap A_j = \emptyset$ for $i \neq j$,
 - $\bigcup A_i = S$
- Equivalence classes form a partition of the set.

Key Properties

- Every equivalence relation induces a partition of the set.
- Every partition defines an equivalence relation.
- Thus, equivalence relations $\leftrightarrow$ partitions.

Conclusion

- Equivalence relations group elements by “sameness”.
- They provide a basis for many mathematical constructions:
 - Integers mod n
 - Rational numbers from integer pairs
 - Real numbers via Cauchy sequences

Definition of a Function

A function f from a set D to a set R is a rule that assigns each $x \in D$ a unique element $f(x) \in R$.

- D is the domain of f
- R is the codomain
- $f(D)$ is the range

Definition of a Function

A function f from a set D to a set R is a rule that assigns each $x \in D$ a unique element $f(x) \in R$.

- D is the domain of f
- R is the codomain
- $f(D)$ is the range

In real analysis, we often consider $f : \mathbb{R} \rightarrow \mathbb{R}$.

Graph of a Function

- The **graph** of f is the set:

$$G(f) = \{(x, f(x)) : x \in D\}$$

- Visual representation of function behavior
- Important for analyzing continuity, limits, etc.

Graph of a Function

- The **graph** of f is the set:

$$G(f) = \{(x, f(x)) : x \in D\}$$

- Visual representation of function behavior
- Important for analyzing continuity, limits, etc.

Basic Geometric Properties

- **Intercepts:**

- x -intercepts: $f(x) = 0$
- y -intercept: $f(0)$ (if defined)

- **Symmetry:**

- Even: $f(-x) = f(x)$
- Odd: $f(-x) = -f(x)$

Monotonicity

- f is **increasing** on I if $x < y \Rightarrow f(x) \leq f(y)$
- **Strictly increasing** if $f(x) < f(y)$
- Similar definitions for decreasing functions

Monotonicity

- f is **increasing** on I if $x < y \Rightarrow f(x) \leq f(y)$
- **Strictly increasing** if $f(x) < f(y)$
- Similar definitions for decreasing functions

Important in:

- Inverse functions
- Interval analysis

Piecewise Functions

Some functions are defined differently on different parts of the domain:

$$f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ 2x + 1 & \text{if } x \geq 0 \end{cases}$$

Piecewise Functions

Some functions are defined differently on different parts of the domain:

$$f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ 2x + 1 & \text{if } x \geq 0 \end{cases}$$

Geometrically: function's graph has different shapes on different intervals.

Conclusion

- Functions can be visualized as graphs in $\mathbb{R}^2$
- Geometric properties include intercepts, symmetry, monotonicity
- Understanding these is crucial for deeper topics in calculus